

Question number	Answer	Additional guidance	Mark
3(a)	A description that includes the following points: - contain {deoxyribose / pentose / 5 carbon sugar}, phosphate and bases (1) (mononucleotides / bases) held together by hydrogen bonds (1) - between {complementary bases / named example} (1)	NB ignore statements clearly relating to RNA ACCEPT pair contains purine and pyrimidine / (mono)nucleotide contains either purine or pyrimidine phosphate group / PO_4^{3-} DO NOT ACCEPT P / Pi / wrong formulae IGNORE stated number of H bonds ACCEPT A / T / C / G IGNORE descriptions of mononucleotides joined by phosphodiester bonds in a strand NB points can be awarded from clearly labelled diagrams	(3)

Question number	Answer	Additional guidance	Mark
3(b)	- number of seconds for molecules to replicate calculated (1) 833 (1) OR - number of molecules replicated in 1 hour (1) 833 (1)	(150 million $\div$ 50) 3 000 000 (50 $\times$ 60 $\times$ 60 =) 180 000 Bald answer of 833 = 2 marks Bald answer of 833.3 = 1 mark unless given as a recurring number	(2)

Question number	Answer	Additional guidance	Mark
3(c)(i)	An explanation that includes the following points: • bind to each strand of the DNA (to initiate replication) (1) • credit function of DNA polymerase (1) • so that the DNA can be synthesised in both directions (1)	ACCEPT works at both ends of the bubble IGNORE ref to 3' / 5' e.g. lines up nucleotides (along each strand) forms phosphodiester bonds (between adjacent nucleotides) repairs mistakes in replication IGNORE forms hydrogen bonds between nucleotides	(2)

Question number	Answer	Additional guidance	Mark
3(c)(ii)	An answer that includes the following points: • to speed up the process (of DNA synthesis) (1) • so that S phase {is shorter / lasts 8 hours and not 833 hours} (1) • so that cell division is fast enough (1)	ACCEPT wrong figures implied from a wrong calculation	(3)